

SERVICE PROPOSAL

NPIC QC & Operations System

Phase 2 — Production Operations

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1. Overview

This proposal outlines the scope of Phase 2 of the NPIC digital operations platform. Building on the QC inspection system established in Phase 1 (Modules A - I), Phase 2 focuses on digitizing the production floor — replacing the daily Excel log and paper-based cart records with a real-time, role-controlled system, moving powder blending from manual estimation to formula-driven calculation, and integrating directly with Sage 100 to eliminate manual double-entry.

Phase 2 introduces six modules (A - F):

Module	Name	Purpose
A	Production — Supervisor Data Entry	Batch logging of employee output by supervisors, replacing the Excel daily report
B	Production — Employee Station Mode	Real-time cart-by-cart entry by employees at the machine, with sign-in/out
C	Dry Room Operations	Cart scan-in/out linked to production records from Module B
D	Data Summary & Reporting	Aggregated shift reports, productivity metrics, and data exports
E	Sage 100 Integration	Real-time bi-directional data sync between the NPIC system and Sage 100
F	Powder Blending Management	Formula-driven blending plans and powder pick lists, with two-tier powder inventory across the raw material warehouse and the blending room

2. Current State & Challenges

Area	Current Method	Key Challenges
Production Tracking	Excel daily report (2026 Daily Report Forming Production.xlsx)	Manual entry — each supervisor spends approximately 2 hours per day on data entry; error-prone; no real-time visibility; productivity metrics calculated by hand
Dry Room	Paper-based cart logs	No systematic cart-chain visibility; half-cart continuations are difficult to track across sessions

Area	Current Method	Key Challenges
Reporting & Analytics	Manual compilation from Excel	No consolidated view; extracting shift summaries or productivity trends requires additional manual work
Sage 100 Data Entry	All production records manually re-keyed into Sage 100 by staff	Time-consuming double-entry; errors from transcription; delays between actual production and system records
Powder Blending	Blending quantities are currently estimated manually in Sage 100; there is no separate powder inventory ledger on the blending room side	Sage 100 does not expand formulas — blend ratio consistency depends on individual experience; the blending room tier of inventory does not exist in any system, so powder loses ledger visibility the moment it leaves the raw material warehouse, and issues and actual consumption cannot be evidenced; variance between planned and actual usage cannot be attributed, leaving material loss unanalyzable

3. System Architecture

Phase 2 is built on the same technical stack as Phase 1 and integrates directly into the existing NPIC platform. New modules share the same user authentication, barcode scanning, and export infrastructure.

Role & Access Design

Role	Modules Accessible	MFA Required?
Employee	Module B — Station Mode (sign-in/out, cart entry)	No
Supervisor	Modules A, B, C, D — full entry, review, and reporting; Module F — blending plan generation and review	No (standard operations)
Raw Material Warehouse	Module F — pick list issue confirmation and stock deduction	No
Blending Room	Module F — pick list review and actual-usage entry	No
Manager	All modules + restricted actions (work order close)	Yes — Microsoft Authenticator
Admin	Full system access including Sage 100 integration configuration	Yes — Microsoft Authenticator

Key Design Principles

- **Employee Mode** — Lightweight production-floor interface. Badge scan or employee ID sign-in. No MFA required for daily entries.
- **Supervisor Mode** — Full batch data entry matching the layout of the existing Excel form.
- **Privileged Actions** — Work order closing requires Microsoft Authenticator verification (inherited from Phase 1, Module I).
- **SKU-Configurable Fields** — Required fields (weight vs. quantity, cart tray layers, work order fields) are pre-configured per product SKU.
- **Half-Cart Linking** — Physical cart continuations are bound in the production stage (Module B); all downstream modules inherit the association automatically.
- **Two-Tier Powder Inventory** — Powder is ledgered separately in the raw material warehouse and in the blending room. Issuing from the warehouse is a **transfer**; only the actual-usage entry filed by the blending room counts as **consumption**. The two postings are recorded separately, each backed by its own document.

4. Module Specifications

Module A Production — Supervisor Data Entry

Replaces the existing Excel daily report. Supervisors enter batch records for all workers and carts in a given shift or period. This mode is used by teams that do not have individual barcode scanning stations.

Data Captured

- Employee names and IDs
- Machine assignments
- Work order numbers (工单号)
- Primary material codes — 一级单 (raw material identifiers)
- Cart numbers, tray layers, and quantity or weight per cart
- Shift hours and total labor time

Outputs

- Auto-calculated per-employee productivity metrics (fed into Module D reporting)
- Records linked to work orders for Sage 100 sync via Module E

Module B Production — Employee Station Mode

A streamlined, employee-facing interface for real-time production logging at the machine or line level. Designed for shared tablets or terminals on the production floor. No MFA required for standard entries.

Sign-In / Sign-Out

- Employees sign in before the shift by entering their employee ID or scanning their badge. Multiple employees can sign in simultaneously on the same machine.
- A digital signature is required at sign-out.
- The shift closes when all employees assigned to that line have signed out.

Cart Entry

- On the first cart, the employee inputs the primary work order number(s). Subsequent carts auto-populate from the last entry — a confirmation button must be pressed before proceeding; manual input can override the default at any time.
- One cart can be associated with multiple primary work orders.
- Required fields (weight or quantity, tray layers) are pre-configured per product SKU.
- Cart number and tray layer count are recorded for each entry.

Half-Cart Continuation

- For the first cart of a session, the employee must select one of the following — mandatory, cannot be skipped:
- **Standalone cart** — a new, independent production run.
- **Half-cart (continuing)** — a physical continuation of a previous session's cart. The system links this cart to the prior cart number. All downstream modules (dry room, reporting, Sage 100 sync) inherit this association automatically — no re-selection required anywhere else in the workflow.

Work Order Close — Restricted (MFA Required)

Because actual production output is typically less than the theoretical BOM quantity, a work order close function is provided. This action requires special permission and Microsoft Authenticator verification.

Option	Description
No Remainder	All materials fully consumed. Work order closes cleanly.
Scrap Remainder	Remaining raw material written off. System auto-distributes unused quantity evenly across finished carts for accurate material accounting.
Retain Remainder	Remaining material reserved for a future run. No adjustment to cart records.

Module C Dry Room Operations

Tracks carts through the dry room with production-record linkage. No half-cart re-selection is required — the association established in Module B carries through automatically.

- **Scan-In:** Operator scans the cart barcode on entry. The system retrieves the cart's production record (including any half-cart binding) automatically from the barcode and records the entry timestamp.
- **Scan-Out:** Records exit timestamp when the cart leaves the dry room.

Additional dry room workflow details (redry queue, time/temperature logging) will be confirmed with the NPIC team during the requirements phase.

Module D Data Summary & Reporting

Aggregates all production data recorded in Modules A and B into structured reports and exports, replacing the manual compilation currently done from the Excel file.

Shift & Daily Reports

- Auto-generated shift summary: total carts, total quantity/weight, work hours, and headcount per shift
- Daily production summary across all machines and work orders for a given date
- Per-work-order completion report: planned vs. actual quantities, material reconciliation status

Productivity Reports

- Per-employee output (carts completed, weight/quantity produced, hours worked) over any date range
- Machine utilization: active hours vs. idle time per line
- Trend view: week-over-week or month-over-month output by product or work order

Data Export

- All reports exportable by date range or product SKU
- Export formats: Excel and PDF
- Leverages the Phase 1 export framework for consistency

Module E Sage 100 Integration

Connects the NPIC system directly to the Sage 100 database for real-time data exchange — eliminating the manual re-entry of work orders, inventory adjustments, and production completions between the two systems.

Data Pulled from Sage 100

Data	Purpose in NPIC System
Work orders (工单)	Auto-populate work order numbers in Modules A and B — no manual keying required
Item master / BOM (一级单)	Sync SKU configurations, product descriptions, and raw material codes used throughout the system
Customer POs	Link work orders to customer purchase orders for end-to-end traceability
Inventory levels	Display real-time raw material availability when opening a new work order
Powder stock balances and every new receipt entry	Serve as the inventory basis for Module F blending plans; each new raw material receipt posted in Sage 100 syncs to the NPIC system so plans always reflect current stock

Data Pushed to Sage 100

Data	Trigger
Production completions (finished goods receipts)	Automatically posted when a work order is closed in Module B
Raw material consumption	Material quantities consumed (from 一级单 reconciliation at close) posted to Sage 100 inventory
Lot numbers	Production lot assignments synced to Sage 100 for end-to-end traceability
Cart / quantity records	Final production quantities from Modules A/B posted to Sage 100 as they are confirmed
Actual blending consumption (via BOM)	Once the blending room files actual usage in Module F, the consumed quantity of each powder is written back to Sage 100 through the BOM

Sync Design

- **Real-time pull:** Work order and item master data is fetched from Sage 100 on demand when a user opens a new record, ensuring the latest data is always used.
- **Triggered push:** Production completions and inventory adjustments are posted to Sage 100 at defined workflow milestones (work order close) — not on a polling schedule — minimizing conflicts.

- **Error handling:** If a push to Sage 100 fails, the system queues the record and retries automatically. Supervisors are notified of any items requiring manual resolution.

Module F Powder Blending Management

Replaces the manual estimation done today in Sage 100 with a system-generated blending plan. When a work order is opened, the system expands each powder quantity from the SKU's blending formula, issues pick lists to the raw material warehouse and the blending room, and records every transfer and consumption across both powder inventories.

Establishing the Two Powder Inventories

Module F rests on two powder inventory ledgers built in the system. They differ in where their data originates and in how they link to Sage 100:

Inventory	How it is established	Link to Sage 100
Raw material warehouse powder stock	Built from Sage 100 as the source: opening balances and item master imported from Sage 100	Continuously linked — stock balances and every new receipt entry sync from Sage 100 into the NPIC system so blending plans always reflect current stock; deductions from pick list issues are reflected back to Sage 100
Blending room powder stock	Newly created in the system — this tier of inventory does not exist in Sage 100 today and is ledgered here for the first time	Opening balance entered by physical count; thereafter increased by transfers in from warehouse issues and decreased by actual consumption ; consumed quantities written back to Sage 100 via the BOM

The posting timing for both tiers (sync to Sage 100 at issue, or a single write-back once blending is complete) will be settled with NPIC and the Sage 100 owner during the requirements phase.

SKU Blending Formula

- A blending formula is maintained per product SKU. Each formula line records a powder and its percentage share, with version control and effective-date management.
- The formula is the single source of calculation for blending plans. Formula maintenance requires dedicated permission, and plans already issued are unaffected by subsequent formula changes.

Blending Plan Generation

- A **planned weight** is entered for each primary work order at work order creation. The system expands the required quantity of every powder from that SKU's formula.
- The system checks both powder inventories — the **raw material warehouse** and the **blending room** — and builds the plan on the rule: **consume the blending room's existing stock first; draw only the shortfall from the raw material warehouse.**

- The shortfall drawn from the warehouse is assigned to lots automatically on a **first-in-first-out (FIFO)** basis. Lot numbers and per-lot quantities are printed directly on the document, so no on-the-spot judgment is required from the warehouse operator.

Powder Pick List — Two Versions

The first step of a blending plan is the pick list. A single plan produces two documents with different contents, one for each recipient:

Recipient	Document Contents
Raw Material Warehouse	Only the powders to be issued from the warehouse, with quantities and FIFO-assigned lots
Blending Room	The complete material list for the blend — the portion drawn from the warehouse and the portion taken from the blending room's own stock, each labeled separately

Issue & Stock Transfer

- When the warehouse operator confirms **Issued**, the system completes both postings at once: the raw material warehouse is **deducted** by the quantity issued, and the same quantity is **transferred into** the blending room's inventory.
- Powder moved into the blending room remains in inventory and is not yet counted as consumed. No separate goods-receipt confirmation is required from the blending room, keeping floor steps to a minimum.

Actual Usage Entry

- After blending, the blending room enters the **actual quantity used** of each powder on a dedicated page. The system deducts that quantity from the blending room's inventory — this posting is the true consumption.
- **Actual below plan:** the remaining powder stays in the blending room's inventory and can be used directly on a subsequent work order; no return document is required.
- **Actual above plan:** over-issue is not permitted. An additional pick list must be raised, ensuring every stock movement is backed by a document.
- Actual usage is written back to Sage 100 through the BOM (see Module E) and compared line by line against the plan to produce a blending variance record.

Units of measure, the handling when both inventories combined still fall short (block work order creation vs. flag the shortage and proceed), and the pick list print format will be confirmed with the NPIC team during the requirements phase.

5. Implementation Plan

Phase	Activities	Duration
Requirements Finalization	Confirm field definitions, SKU configurations, permission matrix, and Sage 100 data mapping with NPIC team; review existing Excel template; load per-SKU blending formulas and confirm the two-tier powder inventory definition	1 week
Development	Build Modules A - F with role-based access; Sage 100 API integration; connect to Phase 1 authentication and export framework	7 - 9 weeks
Internal Testing	End-to-end system testing using NPIC production data; Sage 100 sync validation in staging environment	2 weeks
Parallel Run	Run new system alongside existing Excel log; compare outputs and resolve discrepancies; validate Sage 100 data accuracy	2 weeks
Training	Train supervisors and employees on Modules A - D; document SOPs for each role	1 week
Go-Live	Full cutover; retire Excel daily report; Sage 100 manual entry discontinued	—

6. Deliverables

- Phase 2 modules (A - F) integrated into the existing NPIC operations platform
- Role-based access control: Employee / Supervisor / Raw Material Warehouse / Blending Room / Manager / Admin
- Microsoft Authenticator MFA gating for work order close
- Auto-generated shift summaries and per-employee productivity reports
- Data export by date range or product (Excel and PDF)
- Sage 100 bi-directional integration: real-time work order and raw material inventory pull; automated production completion, inventory, and blending consumption push
- Powder blending management: SKU formula maintenance, automated blending plans, two-version pick lists, actual-usage entry; establishment of both powder inventory ledgers — raw material warehouse (continuously linked to Sage 100) and blending room (newly created in the system)
- Training documentation for each module and role

7. Expected Benefits

Benefit	Impact
Eliminate manual Excel daily report	~2 hours of supervisor data entry saved per day; real-time production visibility
Real-time cart-level data	Accurate, timestamped records from the production floor the moment each cart is completed
Automated productivity metrics	No manual calculation; shift summaries and per-employee output reports generated instantly
End-to-end cart traceability	Half-cart linking ties production records across sessions; dry room and reporting modules inherit the association automatically
Eliminate Sage 100 double-entry	Work orders auto-populated from Sage 100; production completions and material consumption posted back automatically
Live inventory accuracy in Sage 100	Sage 100 inventory updated at the moment of production close — no end-of-day batch upload or manual reconciliation
Seamless integration with Phase 1	Single platform for QC inspection and production operations; shared authentication, barcode, and export infrastructure
Work-order-level raw-material usage tracking	Actual usage attributed accurately per work order; cross-work-order material mixing eliminated
Accurate cart count & timing tracking	Actual cart count = registered cart count; under-reporting eliminated; 100% output-registration accuracy
Post-production waiting-time logging	Accumulated waiting-time data provides a basis for future process optimization
No manual entry of workstation & labor hours	Workstation and labor-hour data captured automatically, replacing paper/Excel manual entry
Employee cart-level productivity	Computed automatically and accurately per employee × cart — finer-grained, more objective, real time
Work-order-level real-time production tracking	Each work order's progress is visible in real time, supporting more accurate monitoring and planning
No manual filling of per-cart records & reports	Line workers no longer manually fill in per-cart records and statistical reports

Benefit	Impact
Formula-driven blending quantities	Manual estimation eliminated; blend ratio consistency guaranteed by the system, reducing wrong-material and over-blending errors
Visibility of both powder inventories	Warehouse and blending room powder stock visible in real time; issuing is a transfer, blending is the consumption — every movement leaves a trail
Blending variance attributable	Actual powder usage compared line by line against plan per work order, supporting loss control and formula optimization

Detailed Notes

7.1 Eliminate manual Excel daily report. Today, each supervisor manually consolidates the shift's production in Excel and compiles a daily report. Once the system is live, the report is generated automatically from real-time shop-floor capture, with no manual entry. At roughly **2 hours/day** per supervisor, across **10 supervisors × 22 working days**, this saves about **440 labor-hours/month (≈2.5 FTE)**; the daily report shifts from “available next day” to “available in real time.”

7.2 Real-time cart-level data. Each cart generates a timestamped record the moment it is completed — no after-the-fact reconstruction from memory. Backfilling approaches **0**; **per-cart traceability reaches 100%**, with timestamps accurate to the minute.

7.3 Work-order-level raw-material usage tracking. Actual issuance and consumption are recorded **per work order**. Current pain point: material from one work order is often used on another, so actual usage cannot be attributed accurately. Binding material to the work order eliminates the distortion caused by **cross-work-order material mixing**, allowing each order's actual usage / yield to be computed accurately — supporting cost accounting and over-consumption detection.

7.4 Accurate cart count & timing tracking. Each cart's actual output quantity plus its start/finish timestamps are captured in real time. Current pain point: employees may **deliberately omit or under-report carts**, so the actual number produced does not match the registered number. With real-time counting the two match. **Actual cart count = registered cart count**, under-reporting drops to zero, output-registration accuracy rises to 100%, and an **average takt time (minutes/cart)** can be derived for capacity estimation and scheduling.

7.5 Post-production waiting-time logging. The **waiting time after each cart finishes production** (e.g., waiting to enter the next step / the drying room) is recorded, building up data that provides a basis for future process optimization and efficiency improvement. The **average waiting time (minutes/cart)** can be measured and used as a quantified baseline and tracking metric for improvement.

7.6 No manual entry of workstation & labor hours. Workstation assignment and working hours are captured by the system, replacing paper/Excel manual entry. The labor-hour saving here is **already included in the 7.1 daily-report saving and is not double-counted**; the value lies in

improved completeness and accuracy of the labor-hour / workstation data (eliminating missing and incorrect entries).

7.7 Employee cart-level productivity. Productivity is computed automatically and accurately per **employee × cart** (carts/shift, units/labor-hour) — finer-grained, more objective, and real-time than the current approach. Data granularity reaches “person × cart” ; **productivity dispersion** across teams/individuals can be compared, supporting performance evaluation and staffing decisions. (An objective baseline already exists; this item makes the data more accurate and more granular, not something created from nothing.)

7.8 Work-order-level real-time production tracking. The production progress/status of **each work order** is tracked in real time, so planning and production directors can **monitor and plan more accurately** based on actual performance (scheduling, capacity, delivery dates). Work-order progress is visible in real time (100% coverage), shifting from “after-the-fact roll-up” to “real-time command.”

7.9 No manual filling of per-cart records & reports (line workers). Today, line workers must manually fill in each cart's production record and statistical reports; once the system is live, data is captured in real time on the shop floor and workers no longer fill anything in manually. At **1 minute/cart × 6 carts/shift × 3 shifts/day × 40 production lines = 720 minutes/day (12 labor-hours/day)**, across **22 working days**, this saves about **264 labor-hours/month (≈ 1.5 FTE)**.

7.10 Digitized powder blending. Today, blending quantities are estimated manually in Sage 100, which does not expand formulas, and there is no separate powder inventory ledger on the blending room side — powder loses ledger visibility the moment it leaves the raw material warehouse. Once the system is live, blending quantities are expanded automatically from the SKU formula against the planned weight; pick lists are issued in two versions with FIFO-assigned lots; and issuing (transfer) and blending (consumption) are posted separately. Blend ratio consistency no longer depends on individual experience; **book-to-physical accuracy** improves across both powder inventories, with every issue and consumption traceable to a specific work order and lot; and **actual vs. planned powder usage** can be compared line by line per work order, establishing a loss baseline that supports formula optimization and over-consumption detection.